

Module 2, Lesson 3

AI 101 for Mechanical Engineers

**When ML is
worth considering**

Course Structure

1.	Comparison Baselines
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2.	Model Classification
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3.	Hybrid-Model Implementation
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4.	Project Validation
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Section 1: Comparison Baselines

Purpose: Establish criteria for prioritizing simple equations over machine learning complexity.

Prioritize Simple Analytical Baselines



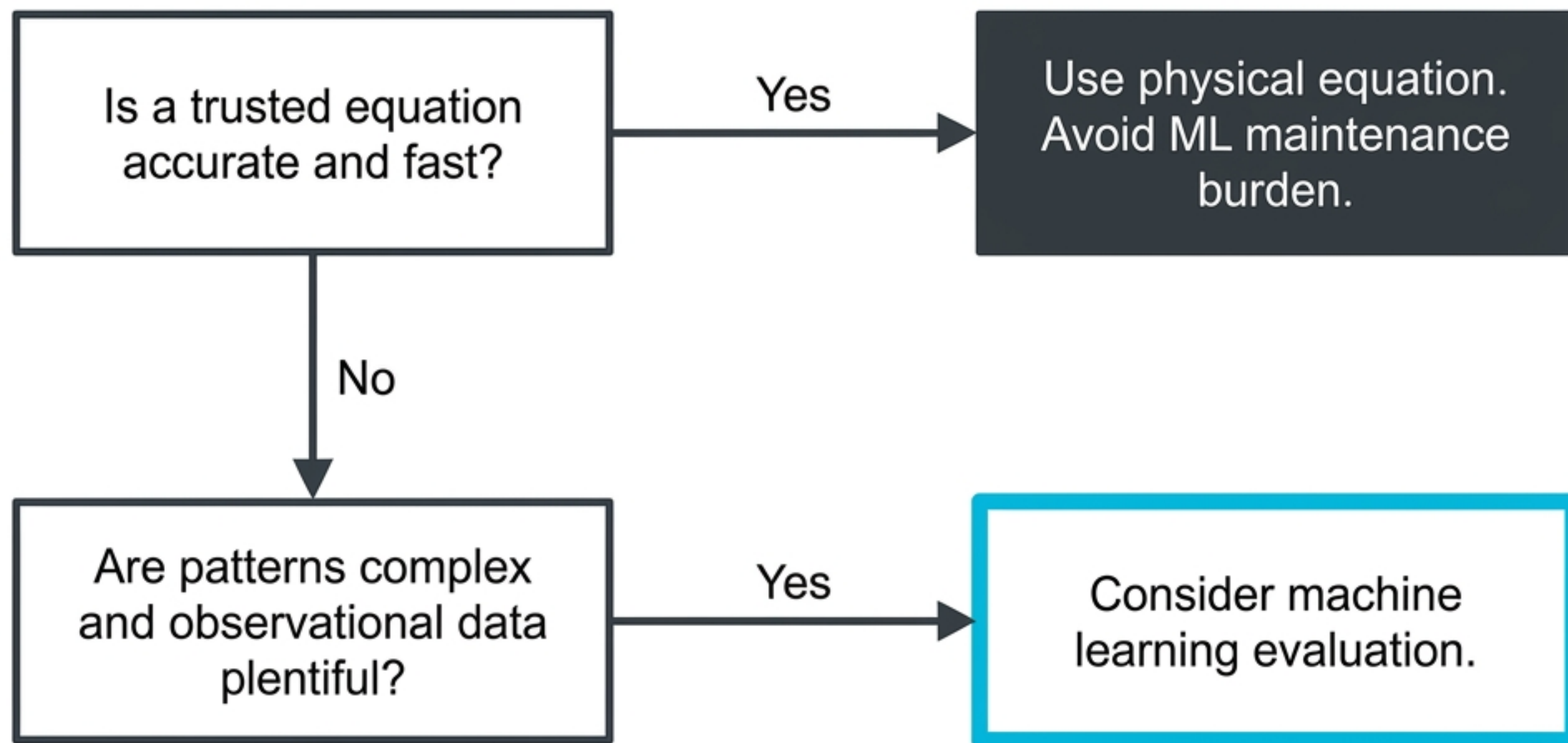
Begin analysis with the simplest adequate engineering baseline.

Reliable equations negate the maintenance burden of machine learning.

Justify machine learning only when analytical accuracy is insufficient.

Consider machine learning for complex patterns or costly simulations.

Analytical Restraint Flowchart



Section 2: Model Classification

Purpose: Formally categorize engineering models by fundamental mechanism and required inputs.

Core Engineering Model Architectures

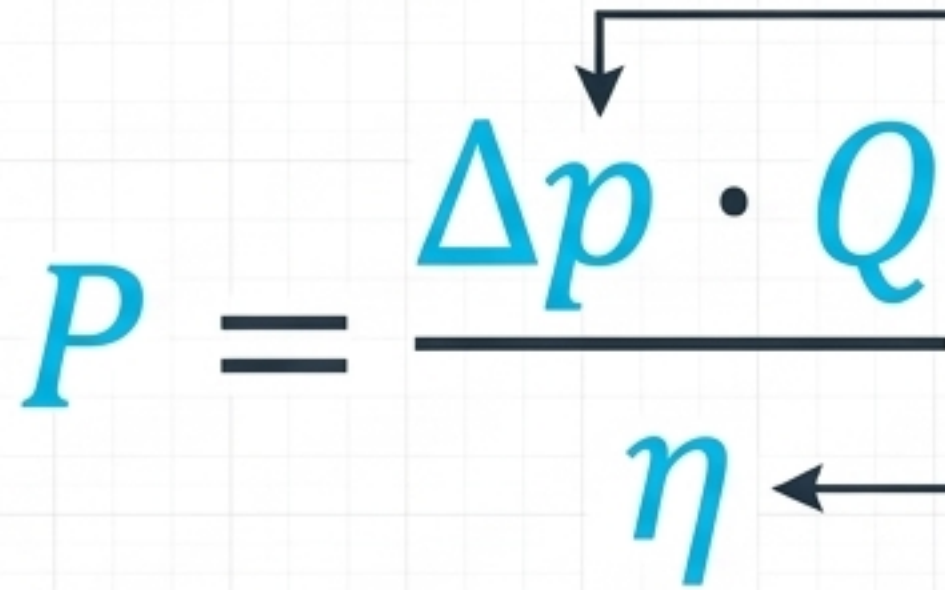
Model Type	Fundamental Mechanism	Required Inputs	Application Example
Physics-based	Physical laws and relationships	Known variables and assumptions	Hydraulic power equation calculation
Data-driven	Inferences from observed output	Large observational datasets	Operating record power predictions
Hybrid model	Combined physical and learned structures	Analytical assumptions plus empirical data	Datasheet residual error correction

Section 3: Hybrid-Model Implementation

Purpose: Prove mathematical integration of machine learning using residual calculation.

Establishing Theoretical Output

Define physical relationships and stated operating assumptions initially.

$$P = \frac{\Delta p \cdot Q}{\eta}$$


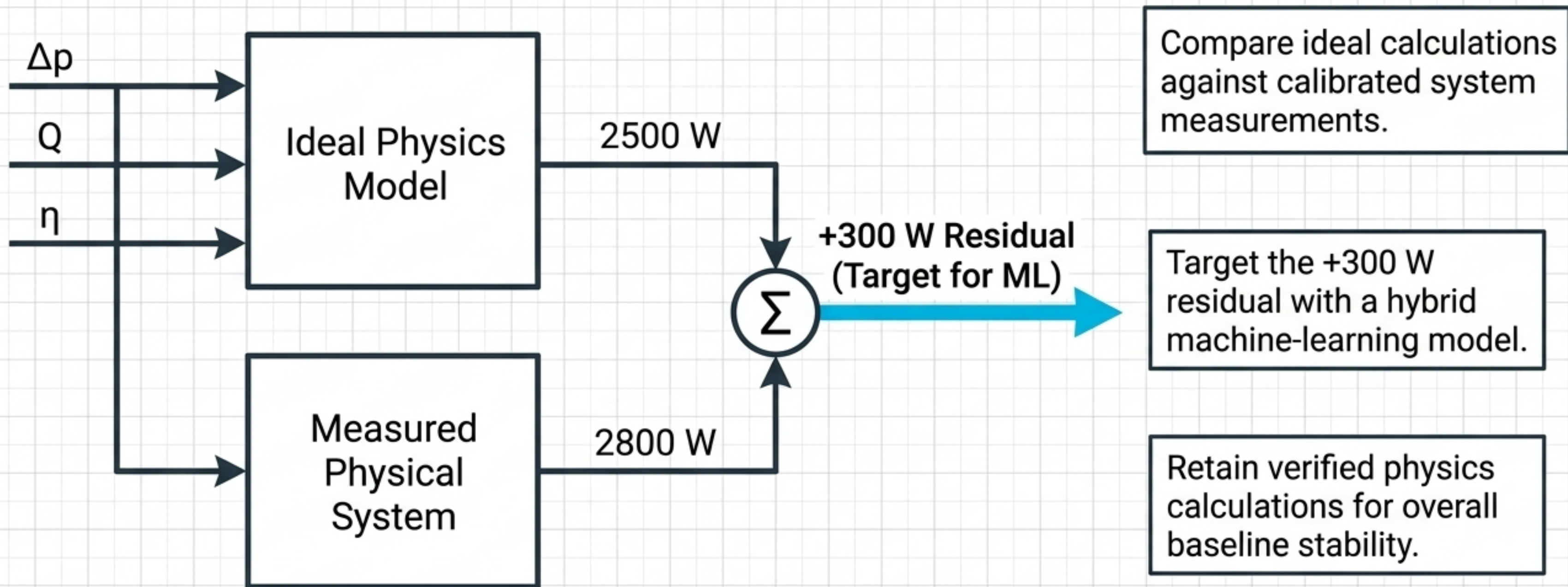
Input pressure differential:
 $\Delta p = 100 \text{ kPa}$.

Input volumetric flow rate:
 $Q = 0.02 \text{ m}^3/\text{s}$.

Input mechanical efficiency:
 $\eta = 0.8$.

Ideal analytical relationship
yields 2500 W input power.

Isolating the Machine Learning Target



Section 4: Project Validation

Purpose: Define strict pre-project readiness criteria for machine learning implementation.

Machine Learning Readiness Pillars



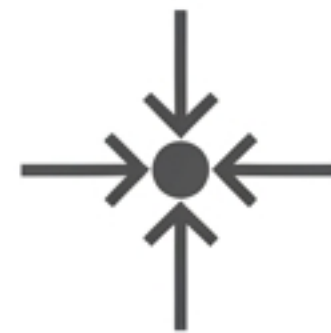
Purpose

Defined engineering decision required.



Baseline

Adequate simple comparison established.



Inputs

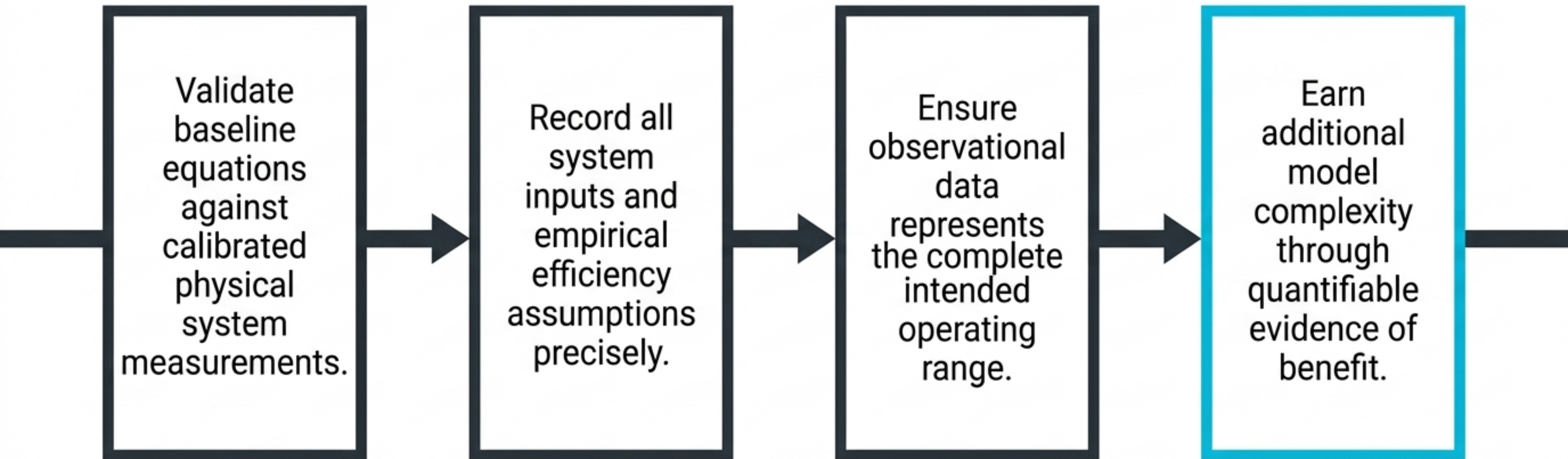
Variables demonstrably available at use time.



Coverage

Data represents intended operating conditions.

Baseline and Assumption Validation



Key Points

1	Prioritize trusted analytical equations over unproven machine learning complexity.
2	Classify models strictly by fundamental mechanisms and required empirical inputs.
3	Isolate system residuals mathematically to justify hybrid machine learning integration.
4	Validate projects using strict engineering criteria for baseline comparison readiness.